

Serum Albumin Binding of Structurally Diverse Neutral Organic Compounds: Data and Models

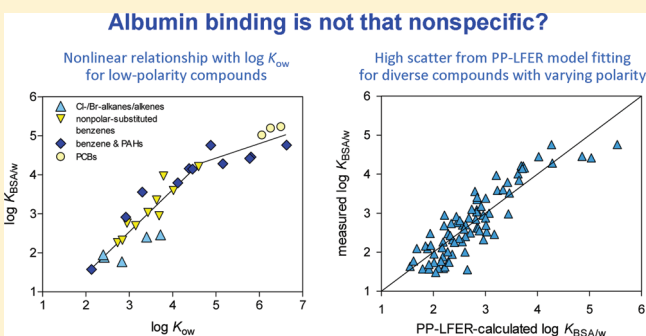
Satoshi Endo^{*,†} and Kai-Uwe Goss^{†,‡}

[†]Department of Analytical Environmental Chemistry, UFZ-Helmholtz Centre for Environmental Research, Permoserstrasse 15, D-04318 Leipzig, Germany

[‡]Institute of Chemistry, University of Halle-Wittenberg, Kurt-Mothes-Strasse 2, D-06120 Halle, Germany

 Supporting Information

ABSTRACT: Binding to serum albumin has a strong influence on freely dissolved, unbound concentrations of chemicals *in vivo* and *in vitro*. For neutral organic solutes, previous studies have suggested a log–log correlation between the albumin–water partition coefficient and the octanol–water partition coefficient (K_{ow}) and postulated highly nonspecific binding that is mechanistically analogous to dissolution into solvents. These relationships and concepts were further explored in this study. Bovine serum albumin (BSA)–water partition coefficients ($K_{BSA/w}$) were measured for 83 structurally diverse neutral organic chemicals in consistent experimental conditions. The correlation between $\log K_{BSA/w}$ and $\log K_{ow}$ was moderate, with $R^2 = 0.76$ and $SD = 0.43$. The $\log K_{BSA/w}$ of low-polarity compounds including a series of chlorobenzenes and polycyclic aromatic hydrocarbons increased with $\log K_{ow}$ linearly up to $\log K_{ow} = 4–5$, but then the linear relationship apparently broke off, and the increase became gradual. The fitting of polyparameter linear free energy relationship models with five solute descriptors was just comparable to that of the $\log K_{ow}$ model ($R^2 = 0.78–0.79$, $SD = 0.41–0.42$); the relatively high SD obtained suggests that solvent dissolution models are not capable of modeling albumin binding accurately. A size limitation of the binding site(s) of albumin is suggested as a possible reason for the high SD . An equilibrium distribution model indicates that serum albumin generally has high contributions to the binding in the serum of polar compounds and relatively small low-polarity compounds, whereas albumin binding for large low-polarity compounds is outcompeted by the strong partitioning into lipids due to low relative affinity of albumin for these compounds.



INTRODUCTION

Ligand binding by serum albumin is of paramount importance in evaluating the toxicology of contaminants. The extent of binding to albumin is directly related to the unbound (or freely dissolved) fraction of chemicals in blood, which is the fraction considered available for biological processes such as membrane permeation, metabolic reactions, and toxic actions. Moreover, culture media used for *in vitro* cell toxicity assays commonly contain serum from fetal or newborn calves. Binding of test chemicals by medium components, serum albumin among others, can significantly reduce the freely dissolved concentrations that the cells are exposed to and has to be considered for *in vitro*–*in vivo* extrapolations.^{1–4} Furthermore, serum albumin (in particular, bovine serum albumin (BSA)) is often regarded as a generic protein to investigate the extent of protein–chemical interactions.⁵ Partitioning of chemicals to the protein fraction of tissues and organs is of general interest to better account for the disposition of chemicals in organisms.⁶ Although it has not been validated that the binding properties of serum albumin are comparable to those of other proteins, serum albumin is

considered as a model protein because binding constants are available abundantly.

Binding to human serum albumin (HSA) has been intensively investigated in pharmaceutical science and industry, as understanding blood protein binding is crucial for successful drug development. Previous studies have identified two primary binding sites and many secondary sites on HSA.^{7–9} The two primary sites are largely nonpolar cavities with characteristic polar functional groups.⁹ A number of quantitative structure–activity relationship (QSAR) models have been proposed for predicting an affinity constant (K_a [M^{-1}]) or equivalently, the albumin–water partition coefficient ($K_{albumin/w}$ [$L_{water}/kg_{albumin}$]) of drug candidates to HSA (see the Appendix for the definitions of K_a and $K_{albumin/w}$ and their interchangeability at low solute concentrations). For example, Colmenarejo et al.¹⁰ measured HSA binding constants of 95 pharmaceuticals using an immobilized HSA column and derived a multivariate nonlinear model. Using

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the same data set, Wichmann et al.¹¹ developed a linear regression equation with five COSMO-RS σ -moment descriptors. Kratochwil et al.¹² compiled HSA binding constants of 138 compounds from a literature survey and demonstrated that a structure similarity concept can be used to model the HSA binding. More QSAR models can be found in the literature, as reviewed by others.^{13,14} To our knowledge, such extensive modeling efforts have not been made for serum albumins from other species.

The QSAR models mentioned above were mainly calibrated for albumin binding of pharmaceutical compounds that vary widely with regard to the charge state. For the binding of neutral organic compounds, it was already reported in the late 1960s–early 1970s that $\log K_a$ to BSA appears to be correlated with the $\log$ of the octanol–water partition coefficients (K_{ow}).^{15,16} From the correlation, it was concluded that “binding of neutral molecules to BSA is a very nonspecific process which is not sensitive to the geometry of the substrate,”¹⁶ analogously to the dissolution of chemicals to organic solvent. Such a correlation was recently reported again.⁶ However, the correlation between $\log K_{ow}$ and $\log K_a$ (or $\log K_{albumin/w}$) has been evaluated with limited data and the data used were sometimes measured in a high concentration range where the binding sites could have been saturated. Thus, the extent of this relationship should be re-evaluated with a more consistent and diverse set of data. Moreover, if the albumin binding for neutral compounds is truly nonspecific and mechanistically comparable to the dissolution to organic solvents, it is anticipated that a polyparameter linear free energy relationship (PP-LFER) model fits better to the experimental data of K_a and $K_{albumin/w}$ than a K_{ow} regression because the PP-LFER can more accurately describe the molecular interactions relevant for nonspecific partitioning processes.

The purposes of this study are to determine BSA–water partition coefficients ($K_{BSA/w}$) for a wide variety of neutral organic compounds in consistent experimental conditions, to evaluate the simple regression model with $\log K_{ow}$ and the PP-LFER model to estimate binding constants to BSA, and to evaluate relative contributions of serum albumin to the bound fraction in the serum of diverse compounds.

MATERIALS AND METHODS

Materials. Chemicals were purchased from different providers. Lyophilized powder of BSA (>98%, heat-shocked, principally fatty acid free) was purchased from Supelco. Water was purified with a Milli-Q A10 Ultrapure Water Purification System (Millipore, Billerica, MA). A phosphate buffer solution containing 150 mM NaCl and 10 mM phosphate (pH 7.40) was used to dissolve BSA. Glass microfibers coated with either polyacrylate (PA; 36 μ m thick) or poly(dimethylsiloxane) (PDMS; 30 μ m thick) were produced by Polymicro Technologies Inc. (Phoenix, AZ). These fibers were cut into 40 mm pieces and cleaned in methanol for >24 h. Methanol was replaced at least three times. The cleaned fibers were dried in air overnight and stored in vials.

Batch Experiments for the Determination of BSA–Water Partition Coefficients $K_{BSA/w}$. To determine the $K_{BSA/w}$ of chemicals with varying properties, two methods were adopted: headspace sampling and fiber extraction methods. In both methods, the third phase (i.e., the gas or fiber phase) was sampled and analyzed for target chemicals to measure the freely dissolved concentrations in the BSA solution.

Headspace Sampling Method. Headspace measurement is useful to determine the freely dissolved concentration of volatile compounds, as the concentration in the headspace is proportional to the free

concentration in the aqueous phase.^{17,18} BSA solution (0.1–1 w/v %) was pipetted into 10-mL crimp-top vials and was spiked with a methanol stock solution containing 3–6 test solutes. The vials were immediately closed with PTFE- or aluminum-lined silicone septa. Methanol content did not exceed 0.2 v/v %. The vials were shaken in a water bath at 160 strokes/min (GLS 400, Grant Instruments, UK) at 37 °C for 24 h. After equilibration, the headspace was sampled with a gastight syringe and measured with a GC/MS (7890A/5975C, Agilent Technologies, Inc., Wilmington, DE). The details of the headspace–GC/MS measurements are described elsewhere.¹⁹ Isoflurane, enflurane, halothane, and methoxyflurane were measured with GC/ECD (HP 6890, Hewlett-Packard) instead of GC/MS. For GC/ECD measurements, 100 μ L of headspace was manually injected to the GC, with a split ratio of 20:1, and the chemicals were separated with an HP-1 column (30 m $\times$ 0.32 mm i.d., 4 μ m film thickness; Agilent Technologies). In all cases, calibration standard solutions were prepared in buffer solutions and were measured along with the spiked BSA solutions. The bound concentration to BSA was determined based on the solute mass balance. The solute mass fraction in the headspace was considered in the mass balance calculation using the air–water partition coefficients of solutes. Separate spiked BSA samples were prepared in the same manner as the test batches and were liquid–liquid extracted, followed by GC/MS analysis to test mass conservation of test chemicals over the shaking periods.

Fiber Extraction Method. Compounds that are not volatile enough to be measurable in the headspace were measured with a passive fiber extraction method. In analogy to the headspace phase explained above, the fiber phase serves as a mediator for measurements of the unbound concentration in the BSA solution.^{20–22} A glass test tube with a glass stopper received 1–3 pieces of 40 mm fiber, 4 or 5 mL of a 0.1–1 w/v % BSA solution, and up to 10 μ L of a methanolic stock solution of 3–5 test chemicals. A shaking time of 24–48 h was given for the fiber–BSA–water three phase equilibrium. The fiber was then retrieved, cleaned with tissue, inserted into a 1.5 mL vial, and extracted with 200–500 μ L of ethyl acetate or cyclohexane. Complete extraction (97% in the worst case) by the selected solvent was checked for each compound using PP-LFER-estimated partition coefficients into PA, ethyl acetate, and cyclohexane. The extracts were analyzed with the GC/MS, as described in ref 23. From the determined concentration in the fiber and the known fiber–water partition coefficient (K_{fw}), the unbound concentration in the BSA solution was obtained. The K_{fw} values for PA used were reported previously,²³ and the K_{fw} values for PDMS were determined in this study (see section SI-1 in Supporting Information for more information on K_{fw} determination). The concentration bound to BSA was determined based on the mass balance. An aliquot of the BSA solution after equilibrium was liquid–liquid extracted and analyzed to check the mass conservation of test compounds over the shaking period. Liquid–liquid extraction was conducted in a 1.5-mL vial, which was shaken only gently to avoid the formation of the foamy layer.

Experimental conditions for all test compounds are summarized in section SI-2 (Supporting Information). Additional experiments were conducted to determine equilibration times, binding isotherms, and influences of methanol on the results (see the Results section). In both experimental methods, four to five replicate vials were prepared. All chemicals used were >99% neutral at pH 7.4. Assuming 1:1 binding, the bound-to-total mole ratio of BSA (r) was calculated.

$$r = [\text{BSA}_{\text{bound}}]/[\text{BSA}_{\text{total}}] \quad (1)$$

$[\text{BSA}_{\text{bound}}]$ is the molar concentration of BSA that is bound to the solute and $[\text{BSA}_{\text{total}}]$ is the total (i.e., bound + unbound) molar concentration of BSA in the solution. In the experiments, the ratio r was kept low (<0.05) to avoid the saturation of binding sites. Only one concentration level was examined, assuming the concentration independence of $K_{BSA/w}$.²⁴ Only $K_{BSA/w}$ values from the experiments in which at least 20% of the added solute mass was bound to BSA were considered.

Recovery was calculated as follows:

$$\text{recovery} = M_{\text{solution}} / (M_{\text{total}} - M_{\text{fiber}}) \quad (2)$$

where M_{solution} , M_{total} , and M_{fiber} are the solute mass in the BSA solution (and in the headspace for volatile compounds) measured by solvent extraction, the solute mass added to the system, and the solute mass measured in the fiber, respectively. NaN_3 (200 mg/L) was added to the buffer solution in the later stage of our experiments to inhibit microbial activities because some compounds (e.g., alcohols) showed a relatively low recovery in earlier experiments. The influence of 200 mg/L NaN_3 on $K_{\text{BSA/w}}$ was assumed negligible, as confirmed by separate experiments for selected compounds (data not shown).

Literature Data for Albumin–Water Partition Coefficients. Experimental binding constants of serum albumin reported for neutral compounds were evaluated and used for the validation of models. To assemble a consistent data set, we set several criteria. If binding constants for more than one binding site were reported, only the primary constant was taken, as it characterizes the binding at low solute concentrations. Loading of chemicals on albumin was checked, and the data from measurements in only high loading conditions (i.e., $r > 0.2$) were not selected. Only neutral compounds (>99% neutral at solution pH) were considered. If the bound concentration was indirectly obtained from subtraction of the total mass by the unbound mass of the solute, at least 20% had to be bound. All reported values were converted to $K_{\text{BSA/w}}$ as explained in the Appendix. Previous data collections^{12,25} were referred to, but the source articles therein were re-evaluated on the basis of the criteria above. In the end, $K_{\text{BSA/w}}$ values for 46 compounds^{21,22,24,26–31} were compiled. Because many of the literature data for BSA are for simple, monofunctional aliphatic compounds (e.g., ketones, esters, and alcohols), $K_{\text{HSA/w}}$ data for 10 compounds that are more structurally complex^{20,32–41} were additionally considered. The literature data collected are listed in section SI-3 (Supporting Information). If more than one data is available for one compound, the log-mean value is used in the following discussion. Experimental temperature for the literature data was 20–37 °C.

RESULTS AND DISCUSSION

Equilibrium Time. Time-series experiments were conducted with two compounds for the headspace method and eight compounds for the fiber extraction method. In all cases, an apparent equilibrium was attained within 24 h (Figures S2, S3 in section SI-4, Supporting Information). Even the most hydrophobic compound measured here (benzo[ghi]perylene) achieved the three-phase equilibrium between water, fiber, and BSA within 24 h. Although fiber–water partitioning equilibrium of hydrophobic compounds in the absence of BSA may need a longer time period,^{23,42} the presence of BSA accelerated the equilibration process, as has been reported by others.²¹

Binding Isotherms. Isotherms to BSA were determined for naphthalene and di-*n*-pentyl ether. The isotherms were extensively linear from the bound-to-total albumin mole ratio r of ~0.001 up to 0.3 (Figure S4 in section SI-5, Supporting Information). Naphthalene exhibited saturation behavior at $r \sim 1$, while data at $r > 0.3$ were not measured for di-*n*-pentyl ether. High linearity in $r < 0.3$ has been reported for other compounds too.^{26,36,41} These results indicate that $K_{\text{BSA/w}}$ is constant and that a constant $K_{\text{BSA/w}}$ value can represent the binding behavior in low loading conditions. The linear isotherms also suggest no competition between solutes in the low concentration range.

Influence of the BSA Concentration. It has been a controversial issue whether fouling of albumin on fiber influences the result of partition coefficient determination.^{21,43,44} In this work,

Table 1. BSA–Water Partition Coefficients, $K_{\text{BSA/w}}$ [L/kg], Measured at Low Solute Loadings on BSA Using a Buffer Solution (10 mM Phosphate and 150 mM NaCl)^a

compd	log $K_{\text{BSA/w}}$ (37 °C)	SD
<i>n</i> -hexane	3.09	0.05
<i>n</i> -heptane	3.59	0.03
<i>n</i> -octane	4.01	0.02
<i>n</i> -nonane	4.45	0.00
cyclohexane	2.01	0.06
cycloheptane	2.52	0.01
cyclooctane	2.98	0.00
1-nonene	4.22	0.00
1-heptyne	2.49	0.01
1-chlorooctane	3.85	0.02
tetrachloromethane	1.77	0.01
trichloroethene	1.88	0.02
tetrachloroethene	2.40	0.01
tribromomethane	1.95	0.02
γ -hexachlorocyclohexane	2.46	0.01
isoflurane	1.58	0.08
enflurane	1.59	0.05
halothane	1.62	0.04
methoxyflurane	1.77	0.01
di- <i>n</i> -butyl ether	2.01	0.05
di- <i>n</i> -pentyl ether	3.00	0.02
2-octanone	2.09	0.04
2-nonanone	2.48	0.02
2-decanone	2.88	0.02
1-nitrooctane	3.38	0.04
tri- <i>n</i> -butyl phosphate	2.47	0.03
benzene	1.58	0.01
toluene	2.26	0.04
ethylbenzene	2.70	0.03
<i>n</i> -propylbenzene	2.95	0.03
styrene	2.76	0.03
chlorobenzene	2.32	0.04
1,2-dichlorobenzene	3.03	0.02
1,2,4-trimethylbenzene	3.35	0.02
1,4-dibromobenzene	3.97	0.02
1,2,4-trichlorobenzene	3.60	0.03
1,2,3,4-tetrachlorobenzene	4.21	0.02
hexafluorobenzene	1.55	0.04
methylpentafluorobenzene	2.32	0.02
indene	2.92	0.02
naphthalene	3.56	0.01
dibenzofuran	3.79	0.01
dibenzothiophene	4.16	0.02
phenanthrene	4.15	0.02
fluoranthene	4.28	0.01
pyrene	4.76	0.02
chrysene	4.46	0.03
benzo[<i>b</i>]fluoranthene	4.42	0.03
benzo[<i>ghi</i>]perylene	4.76	0.03
anisole	2.16	0.02
valerophenone	2.70	0.01
benzophenone	2.62	0.02

Table 1. Continued

compd	log $K_{BSA/w}$ (37 °C)	SD
di- <i>n</i> -propyl phthalate	2.84	0.02
2-nitrotoluene	2.12	0.02
2,4-dinitrotoluene	1.73	0.03
1-nitronaphthalene	3.17	0.01
4-nitroanisole	2.48	0.01
<i>N,N</i> -diethylaniline	2.27	0.02
1-hexanol	1.64	0.06
1-heptanol	2.18	0.02
1-octanol	2.74	0.02
1-nonanol	3.10	0.01
4-ethyl-3-hexanol	1.48	0.07
4-chlorobenzyl alcohol	2.10	0.02
4- <i>n</i> -propylphenol	2.59	0.01
2-phenylphenol	2.62	0.02
4-fluorophenol	1.57	0.06
3-chlorophenol	2.35	0.01
4-chlorophenol	2.43	0.03
4-bromophenol	2.81	0.02
4-iodophenol	3.41	0.01
bisphenol A	2.88	0.03
4-nitroaniline	1.69	0.15
2-chloroaniline	1.95	0.04
4-iodoaniline	2.95	0.02
4-aminobiphenyl	2.55	0.01
indole	2.25	0.00
carbazole	3.52	0.01
metolachlor	1.74	0.03
atrazine	1.77	0.08
diazepam	2.68	0.02
estrone	2.69	0.04
endosulfan α	3.24	0.08

^a Mean and standard deviations of 3–12 measurements are presented. An extended version of this table with more information is provided in section SI-2, Supporting Information.

$K_{BSA/w}$ values for 8 compounds (including four PAHs) were determined at several BSA concentrations between 0.01 and 1 w/v % using either PA or PDMS fiber. The variation of $K_{BSA/w}$ was within 0.02–0.20 log units (Figures S5, S6 in section SI-6, Supporting Information). Only di-*n*-propyl phthalate exhibited a difference of up to 0.46 log units; this relatively large dependence on the BSA concentration could be at least partially due to a minor error in K_{fw} used for this compound. Two compounds showed somewhat increasing trends with the BSA concentration but others rather decreasing trends. Overall, there was no consistent bias that is attributable to the varying BSA concentration in the data, and thus fouling effects were neglected in the following discussion.

Experimental Values of $K_{BSA/w}$. Values of $K_{BSA/w}$ for 83 compounds measured in this study are shown in Table 1. To our knowledge, this is the largest and the most consistent data set that is publicly available for albumin binding of neutral compounds. For most of the compounds in Table 1, good recovery (>90%) was confirmed by solvent extraction (see section SI-2, Supporting Information). Exceptions were pyrene (88%), fluoranthene (79%), benzo[ghi]perylene (81%), endosulfan α (85%). For these compounds, corrections were made in the mass balance calculation. For some

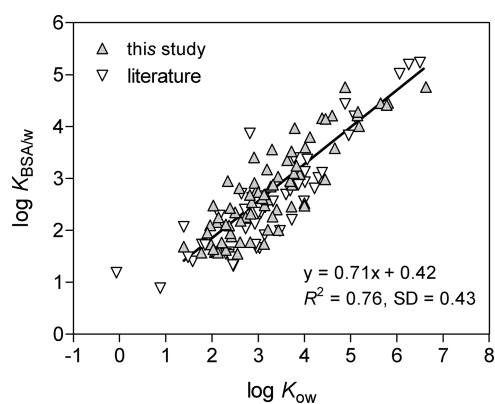


Figure 1. Plot of log $K_{BSA/w}$ against log K_{ow} . The linear regression for the data from this study is shown in the figure. The plotted data from the literature include 9 compounds measured with HSA.

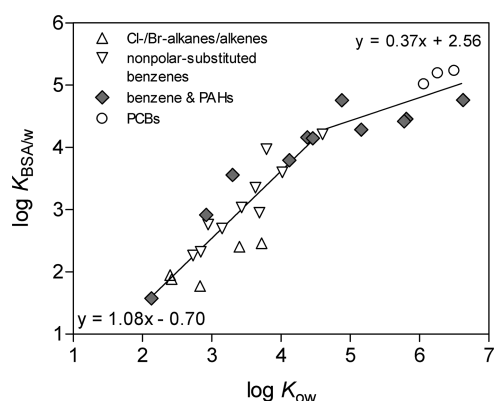


Figure 2. Plot of log $K_{BSA/w}$ against log K_{ow} for low-polarity compounds. The log $K_{BSA/w}$ data for PCBs are from ref 31. Lines indicate linear regressions for log $K_{ow} < 4.5$ and > 4.5 .

volatile compounds, a recovery experiment was unsuccessful or not made. For these, the stability of the compounds was assumed.

Correlation with log K_{ow} . The correlation between log $K_{BSA/w}$ and log K_{ow} is shown in Figure 1. The simple linear regression analysis for the log $K_{BSA/w}$ data measured in this study gives

$$\log K_{BSA/w} = (0.71 \pm 0.05) \log K_{ow} + (0.42 \pm 0.16) \quad (3)$$

$$n = 76, R^2 = 0.76, SD = 0.43$$

The log K_{ow} values used for eq 3 are all experimental data from ref 45 (values for 7 compounds are missing). The data points from the literature distribute similarly in Figure 1 (root mean squared error (rmse) = 0.53, $n = 50$). Indeed, regression analysis for all data from this study and the literature would give a similar equation, log $K_{BSA/w} = 0.71 \log K_{ow} + 0.34$. For literature data, the experimental log K_{ow} values for many compounds were missing and had to be estimated using a PP-LFER model⁴⁶ (see section SI-7 (Supporting Information) for all log K_{ow} values used). Equation 3 is similar to the previously reported regression equations from Vandenberg et al.¹⁶ (log $K_{BSA/w} = 0.65 \log K_{ow} + 0.3$), and deBruyn and Gobas⁶ (log $K_{BSA/w} = 0.57 \log K_{ow} + 0.69$, for log $K_{ow} > 2$). These regression equations from the literature tend to slightly underestimate the values of log $K_{BSA/w}$ from this study in the high

Table 2. Fitting Coefficients and Statistics of PP-LFER Models Obtained for $\log K_{\text{BSA/w}}$ ^a

	<i>c</i>	<i>e</i>	<i>l</i>	<i>s</i>	<i>a</i>	<i>b</i>	<i>v</i>	<i>n</i>	SD	<i>R</i> ²
PP-LFER model (eq 4)	0.14 (0.25)	0.36 (0.12)		−0.26 (0.21)	0.37 (0.24)	−3.23 (0.31)	2.82 (0.25)	82	0.42	0.78
PP-LFER model (eq 5)	0.35 (0.26)		0.28 (0.08)	−0.46 (0.22)	0.20 (0.23)	−3.18 (0.30)	1.84 (0.42)	82	0.41	0.79

^a Experimental data used are listed in Table 1. Values in parentheses are standard errors.

K_{ow} region (Figure S7 in section SI-8, Supporting Information). However, it is more important to note that the correlation between $\log K_{\text{BSA/w}}$ and $\log K_{\text{ow}}$ is relatively weak in general and that there will be errors of ± 1 log units or even larger for a number of compounds, no matter which regression equation is used. Also noteworthy is that, compared to the $\log K_{\text{BSA/w}} - \log K_{\text{ow}}$ plot presented by deBruyn and Gobas,⁶ Figure 1 exhibits a clearly higher degree of scatter. This is explained by the fact that the data set in this study includes a higher number of compounds with higher structural diversity than that in ref 6.

Figure 2 shows the $\log K_{\text{BSA/w}} - \log K_{\text{ow}}$ plot for only low polarity compounds (i.e., compounds that undergo no or only weak H-bond interactions), namely, chlorinated aliphatic compounds, alkylbenzenes, halogenated benzenes, PAHs, and PCBs. Partitioning data for these compounds typically exhibit a linear relationship against $\log K_{\text{ow}}$, as has been shown for partitioning from water into, e.g., poly(dimethylsiloxane),⁴⁷ storage lipids,^{48,49} and natural organic matter.⁵⁰ While $\log K_{\text{BSA/w}}$ does increase with $\log K_{\text{ow}}$ linearly up to $\log K_{\text{ow}} \approx 5$, the slope thereafter becomes gentler, and $\log K_{\text{BSA/w}}$ remains to be around $\log K_{\text{BSA/w}} = 4 - 5$. Note that such a nonlinear plot could also result from experimental artifacts in measured partition coefficients of highly hydrophobic compounds. Indeed, a similar nonlinear relationship was proposed for liposome membrane–water partition coefficients before⁵¹ but was refuted by recent studies.^{52,53} However, the nonlinear trend in Figure 2 is unlikely caused by systematic experimental biases, as the relatively high $K_{\text{BSA/w}}$ values were all measured with polymer-mediated passive sampling (or dosing) methods, which have been proven to be suitable methods for hydrophobic compounds.^{48,54} For estimation purposes, drawing two individual regression lines to high and low K_{ow} compounds may improve the overall fitting, as shown in Figure 2. Indeed, the two regression equations obtained this way are both substantially different from eq 3. However, these regression models are just empirical and should be used only for interpolation of the values for chloroalkanes/alkenes, alkyl-, and halogenated benzenes, PAHs, and PCBs. Also, while there is a clear nonlinear relationship for aromatic hydrocarbons (from benzene to benzo[ghi]perylene), the overall trend in Figure 2 could also be explained, at least partially, by chemical-class specific slopes. Thus, for further interpretation more experimental data are needed.

Polyparameter Linear Free Energy Relationships (PP-LFERs). The data of $K_{\text{BSA/w}}$ were analyzed with two types of PP-LFER model that are slightly different in the parameters used,^{55,56}

$$\log K_{\text{BSA/w}} = c + eE + sS + aA + bB + vV \quad (4)$$

$$\log K_{\text{BSA/w}} = c + lL + sS + aA + bB + vV \quad (5)$$

The notations used for the compound descriptors are *E*, excess molar refraction; *S*, dipolarity/polarizability parameter; *A*, solute H-bond acidity; *B*, solute H-bond basicity; *V*, molar volume; and *L*,

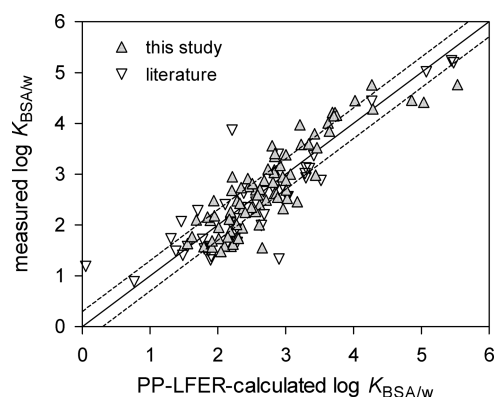


Figure 3. Measured $\log K_{\text{BSA/w}}$ values vs values calculated from a PP-LFER model (eq 4; parameters are given in Table 2). The solid line indicates the 1:1 relationship, and the dashed lines denote 0.3 log unit deviations from the 1:1 line. The data from the literature include 7 data measured with HSA.

the logarithm of the hexadecane–air partition coefficient. The lowercase letters are fitting coefficients and a constant that are calibrated by multiple linear regression analysis based on measured data. These PP-LFERs have been used before to model a number of biphasic partition coefficients. Previous studies have shown that the standard deviation (SD) of estimates is typically 0.1–0.2 for partitioning to homogeneous solvents^{56,57} and amorphous polymer phases,^{23,58} and slightly higher for partitioning to more complex phases such as soil and sediment organic matter (SD = 0.24–0.34)^{50,59,60} and the bilayer phospholipid membrane (SD = 0.28–0.31).⁵³ The values of the descriptors used in this study are given in section SI-7 (Supporting Information). Endosulfan *a* was removed from our data set, and a few pharmaceuticals were removed from the literature data set because no accurate descriptor values were available.

Equations 4 and 5 were fitted to the experimental data from this study (Table 2 and Figure 3). Although the equations appear to capture the general trend as shown in Figure 3, the SDs of estimates (0.41–0.42) were remarkably high for the PP-LFERs (see typical SDs above) and were not better than that of the simple regression with $\log K_{\text{ow}}$ for $\log K_{\text{BSA/w}}$ (i.e., eq 3). Large differences between experimental and fitted $\log K_{\text{BSA/w}}$ were found, for example, γ -hexachlorocyclohexane (+0.71), hexafluorobenzene (+1.10), naphthalene (−0.76), benzo[ghi]perylene (+0.78), and 4-iodoaniline (−0.74) (values are fitted minus experimental). Predictions through the calibrated PP-LFER model (eq 4) for the literature data showed a similar degree of scattering (rmse = 0.51, *n* = 46). Because the PP-LFER models perform generally well for homogeneous solvents, the relatively poor fitting observed suggests that the binding mechanism of BSA cannot be well approximated as partitioning into a solvent, as opposed to the previous postulate.¹⁶

Using the $K_{BSA/w}$ data and the air–water partition coefficients, *n*-alkane-to-cycloalkane distribution coefficient ratios (K_n/K_c) were calculated according to refs 61 and 62. K_n/K_c ratios are indicative of the mode of sorption (i.e., absorption or adsorption); numerous experimental data have shown that K_n/K_c ratios are generally <0.7 for absorption processes into liquids and amorphous polymers, whereas K_n/K_c ratios are 1 or higher for absorption to surfaces.^{61,62} The experimental data here show that the K_n/K_c values of the C_6 , C_7 , and C_8 alkanes are 1.07, 1.05, and 0.97, respectively. These values support the interpretation that binding to BSA is not mechanistically analogous to partitioning into solvents.

Previously, Valko et al.⁶³ calibrated eq 4 using retention data on an immobilized HSA column. The resulting equation, however, systematically underestimates the data measured here as well as those collected from the literature (rmse = 0.86, mean error = -0.79 ± 0.49 , $n = 128$; also see Figure S8 in section SI-9, Supporting Information). Possible reasons for the observed discrepancy include differences between BSA and HSA, different binding properties of immobilized albumin compared to that of native albumin, the calibration data set of ref 63 that is strongly biased to polar aromatic compounds, and the use of an organic modifier for the column eluent in ref 63.

To understand the behavior of low polarity compounds better, we fitted the PP-LFERs to the compounds shown in Figure 2. The aim of this attempt is not to establish a prediction model but simply to test whether any combination of PP-LFER descriptors could explain the BSA binding of, at least, low polarity compounds. The fitting resulted in a SD of 0.40 (Figure S9, section SI-10, Supporting Information), not better than that for the whole data set. Thus, the five PP-LFER descriptors are not sufficient to accurately model $\log K_{BSA/w}$, even only for low polarity compounds. In addition, the residual plot against fitted $\log K_{BSA/w}$ is not uniform (Figure S10, section SI-10, Supporting Information) but concave upward, suggesting that the descriptors could be nonlinearly related to $\log K_{BSA/w}$. This finding also corroborates the nonlinear relationship between $\log K_{BSA/w}$ against $\log K_{ow}$ in Figure 2.

The PP-LFER models assume that $\log K_{BSA/w}$ increases linearly with the size of molecules (as represented by the νV term). However, this assumption might not hold, if binding sites of albumin have limited sizes and large solute molecules cannot optimally interact with albumin molecules. To examine whether such a nonlinear relationship could explain the relatively low fitting of eqs 4 and 5 to the $\log K_{BSA/w}$ data, we attempted to fit two modified PP-LFERs:

$$\log K_{BSA/w} = c + eE + sS + aA + bB + \nu V^{0.5} \quad (6)$$

$$\log K_{BSA/w} = c + eE + sS + aA + bB + \nu_1 V + \nu_2 V^2 \quad (7)$$

In eqs 6 and 7, the size dependency of $\log K_{BSA/w}$ was assumed to be nonlinear and was modeled with an exponent (eq 6) or a second order polynomial expression (eq 7). The exponent was fixed to 0.5 here, as the fitting only slightly depended on the value of the exponent. Equations 6 and 7 were fitted to the data measured in this study (see section SI-11 (Supporting Information) for the resulting coefficients and statistics). The SD of both of eqs 6 and 7 were 0.38, being only slightly better than those of PP-LFERs in the original forms. The resulting models predicted the literature data to the accuracy of rmse = 0.51 and 0.55, respectively,

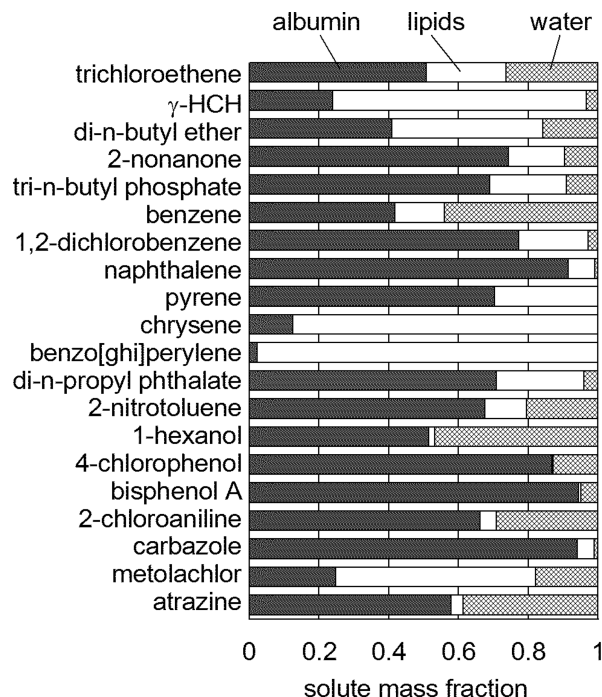


Figure 4. Estimated mass distributions of selected neutral organic solutes in fetal bovine serum. An extended version of this figure is presented in section SI-12, Supporting Information.

which were slightly poorer than of the original equations. These results show that incorporating the nonlinear V term(s) in the model alone does not render the model better performing.

In the previous models, the extent of the polar interactions that the albumin exerts to the solute was assumed to be invariable for any solute. This assumption may not be true if solutes with varying molecular sizes are positioned in distinctly different molecular environments. Therefore, as a next attempt, the data measured in this study were divided into two groups, those with $V < 1.2$ and with $V > 1.2$, and an individual PP-LFER equation was derived for each group (eq 8).

$$\begin{cases} \log K_{BSA/w} = c_1 + e_1E + s_1S + a_1A + b_1B + \nu_1V & (\text{for } V < 1.2) \\ \log K_{BSA/w} = c_2 + e_2E + s_2S + a_2A + b_2B + \nu_2V & (\text{for } V > 1.2) \end{cases} \quad (8)$$

V of 1.2 corresponds approximately to $\log K_{ow}$ of 4 for the low polarity compounds considered in this study. The overall SD was 0.32, indicating better fitting than the models above achieved. Equation 8 uses a greater number of fitting parameters than the original PP-LFERs, but the Akaike Information Criterion still indicates a better model performance of eq 8 than the previous equations. The quality of fitting to relatively small compounds (i.e., $V < 1.2$) was particularly high (SD = 0.28). Predictions to the literature data for small compounds were also of high quality (rmse = 0.33), although there were only 15 compounds that meet the $V < 1.2$ criterion in the literature set. This result suggests that the solvent dissolution model may approximate the binding behavior of albumin, as long as the solute is not larger than the binding pocket(s) of the albumin molecule. In contrast, fitting to the larger compounds ($V > 1.2$) was somewhat poorer (SD = 0.36), and predictions of the literature data were clearly worse (rmse = 0.71). The binding behavior of larger compounds

appears to be more complicated and cannot simply be described by the solvation model. It should be noted that eqs 6–8 use the volume of molecules to describe the size exclusion behavior of the binding sites of albumin. However, not only the volume but also the shape of molecules may have influences on the binding by albumin. That is to say, a bulky molecule may not fit so well to the binding sites as a flexible chain-like molecule of the same volume does. Thus, for further understanding, models that consider molecular interactions on the three-dimensional scale may be needed.

Distribution of Chemicals in Blood Serum. To estimate the influence of albumin binding to the distribution of chemicals in blood serum, we adopted an equilibrium distribution model used by Gülden and Seibert⁶⁴ and Escher et al.²⁴ We assumed that chemicals in serum are distributed between proteins, lipids, and water in equilibrium. Partitioning to serum proteins was represented solely by binding to albumin, which was modeled with the $K_{BSA/w}$ measured in this study. We considered only low concentration cases (i.e., well below saturation of albumin) where $K_{BSA/w}$ is constant. Lipids in serum were represented by storage lipid. Storage lipid–water partition coefficients ($K_{storage\ lipid/w}$ [L/kg]) at 37 °C were estimated using a PP-LFER model⁶⁵ because the PP-LFER has been shown to accurately reproduce experimental partitioning coefficients to storage lipids. Upon the basis of these assumptions, the solute mass fractions in albumin, lipids, and water (denoted as $F_{albumin}$, F_{lipids} , and F_{water} , respectively) can be computed as follows:

$$\begin{aligned} F_{albumin} &= \frac{c_{albumin}K_{BSA/w}}{c_{albumin}K_{BSA/w} + c_{lipids}K_{storage\ lipid/w} + f_{water}} \\ F_{lipids} &= \frac{c_{lipids}K_{storage\ lipid/w}}{c_{albumin}K_{BSA/w} + c_{lipids}K_{storage\ lipid/w} + f_{water}} \\ F_{water} &= \frac{f_{water}}{c_{albumin}K_{BSA/w} + c_{lipids}K_{storage\ lipid/w} + f_{water}} \end{aligned} \quad (9)$$

where $c_{albumin}$ and c_{lipids} represent the albumin and lipid concentrations, respectively, in serum. These were set to 24 and 1.92 g/L, respectively, for fetal bovine serum, and 38 and 7.22 g/L, respectively, for human serum, as done by Gülden et al.⁶⁶ The volume fraction of water (f_{water}) in these sera was assumed to be 0.95. Salting-out by dissolved inorganic salts in serum is unlikely significant for neutral solutes, as shown elsewhere.⁶⁷ Note that the use of $K_{BSA/w}$ to model $K_{HSA/w}$ is an approximation. Also, the use of storage lipid to represent the whole lipid fraction involves uncertainty because lipids in blood are present in the form of lipoproteins, which are made up of triacylglycerides, cholesterol, phospholipids, and various proteins. The purpose here is not to provide the most accurate estimates for individual chemicals but to illustrate the overall trend for diverse chemicals, based on the current best knowledge. Estimations will be improved when more data for partitioning into various blood proteins and lipoproteins from different species become available.

Figure 4 shows the estimated fractional distributions of 20 selected compounds in fetal bovine serum. Figures for 82 compounds in fetal bovine and human sera are provided in section SI-12 (Supporting Information). Albumin, lipids, and water exhibit varying contributions that are highly dependent on the compounds. There is a tendency for polar compounds to distribute primarily to albumin and little to the lipid phase. A notable exception is metolachlor: a 50% mass fraction of this chemical is estimated to partition into lipids in fetal bovine serum because of its relatively low

$K_{BSA/w}$ compared to $K_{storage\ lipid/water}$. Distributions of low polarity compounds differ to a considerably large extent. For relatively small compounds such as alkyl- or halogen-substituted benzenes, the protein fraction is the major fraction. For example, albumin is estimated to retain 90% of the total amounts of naphthalene and 1,4-dibromobenzene in fetal bovine serum. In contrast, the protein contributes only <10% to the binding for the largest of the low-polarity compounds studied (i.e., benzo[*b*]fluoranthene and benzo[*ghi*]perylene), and lipids predominate their distribution. These highly variable distributions of low-polarity chemicals result from the fact that the partition coefficient into lipids increases simply with the molecular size, whereas the partition coefficient to BSA does not, as shown in Figure 2. In human serum, contributions of albumin to the overall binding are generally lower than those in fetal bovine serum (section SI-12, Supporting Information), reflecting the lower protein/lipid ratio in human serum. Note that in high-dose conditions where albumin binding is saturated, partitioning to albumin is weaker, and thus, the relative solute mass in the protein fraction is lower in comparison to the low concentration cases shown here.

This study demonstrated that the log K_{ow} and the PP-LFER models can achieve a model accuracy of 0.4 log unit standard deviations. The assumption of the BSA binding being analogous to the dissolution process into solvents is not suitable for more accurate modeling. For higher accuracy, models may need to capture the interactions between the solute and albumin binding sites in a three-dimensional space. The data and models presented here should also form a basis to evaluate whether BSA can serve as a surrogate for other types of protein.

APPENDIX

The extent of albumin binding is typically reported using the affinity (or association) constant, K_a [M^{-1}], which is defined, e.g., for BSA, as follows:

$$K_a = C_{bound}C_{free}^{-1}[BSA_{free}]^{-1} \quad (10)$$

where C_{bound} is the molar concentration of the bound chemical, C_{free} is the molar concentration of the freely dissolved chemical, and $[BSA_{free}]$ is the unbound concentration of BSA in the solution. For mass balance calculations, it is often more convenient to use the BSA–water partition coefficient defined as follows:

$$\begin{aligned} K_{BSA/w}[L_{water}kg_{BSA}^{-1}] &= q_{BSA}C_{free}^{-1} \\ \text{or} \\ K'_{BSA/w}[L_{water}L_{BSA}^{-1}] &= C_{BSA}C_{free}^{-1} \end{aligned} \quad (11)$$

where q_{BSA} and C_{BSA} are the bound solute moles per unit BSA weight and unit BSA volume, respectively. If solute concentration levels are low and the BSA binding is far below saturation (i.e., $[BSA_{bound}] \ll [BSA_{total}]$), K_a , $K_{BSA/w}$, and $K'_{BSA/w}$ are interchangeable through the relationships as follows:

$$K_a = K_{BSA/w}MW_{BSA} \quad (12)$$

$$K'_{BSA/w} = K_{BSA/w}\phi_{BSA} \quad (13)$$

where MW_{BSA} is the molecular weight of albumin (~ 67 kg/mol), and ϕ_{BSA} is the specific volume of BSA in the solution ($\phi_{BSA} = 0.733$ L/kg was assumed in this work).⁶ Thus, K_a can be obtained from the measured $K_{BSA/w}$ in Table 1 using the relationship, $\log K_a = \log K_{BSA/w} + 1.83$. Throughout this article, we used

$K_{BSA/w}$ to express the extent of albumin binding, but this does not have any mechanistic implication.

■ ASSOCIATED CONTENT

S Supporting Information. More information on PDMS–water partition coefficients, albumin–water partition coefficients, the additional experiments conducted, the values of log K_{ow} and PP-LFER descriptors used, the results of model fitting with eqs 6–8, and estimated mass distribution in fetal bovine and human sera. This material is available free of charge via the Internet at <http://pubs.acs.org>.

■ AUTHOR INFORMATION

Corresponding Author

*Phone: +49 341 235 1818. Fax: +49 341 235 1443. E-mail: satoshi.endo@ufz.de.

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■ ABBREVIATIONS

BSA, bovine serum albumin; HSA, human serum albumin; K_n/K_c , n -alkane-to-cycloalkane distribution coefficient ratio; K_{ow} , octanol–water partition coefficient; PA, polyacrylate; PAH, polycyclic aromatic hydrocarbon; PCB, polychlorinated biphenyl; PDMS, poly(dimethylsiloxane); PP-LFER, polyparameter linear free energy relationship; PTFE, poly(tetrafluoroethylene); rmse, root mean squared error

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